

Phase 9 CVRP Solver Evolution Report

Overview

- Condition count: 4.
- Best held-out condition: phase9_baseline_clarke_wright_savings.

Condition Summary

Condition	Mode	Held-out Feasibility	Held-out Penalized Gap	Held-out Feasible Gap	Held-out Runtime (ms)	Mean Novelty	Mean Complexity
phase9_baseline_nearest_neighbor_constructive	baseline_nearest_neighbor_constructive	1.0	0.273435	0.273435	43.28062	0.0	0.0
phase9_baseline_clarke_wright_savings	baseline_clarke_wright_savings	1.0	0.073766	0.073766	78.47268	0.0	0.0
phase9_baseline_regret_insertion_local_search	baseline_regret_insertion_local_search	1.0	0.466391	0.466391	1492.19562	0.0	0.0
phase9_solver_evolution	solver_evolution	1.0	0.273435	0.273435	137.1848	0.0	0.0

Condition Notes

phase9_baseline_nearest_neighbor_constructive

- Execution mode: baseline_nearest_neighbor_constructive.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 43.28062.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
 - X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
 - X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
 - X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
 - X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]
 - X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

phase9_baseline_clarke_wright_savings

- Execution mode: baseline_clarke_wright_savings.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.064783.

- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.073766.
- Held-out runtime ms: 78.47268.
- Robustness dispersion across held-out structure families: 0.007791.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.108521, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.099285, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.061249, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.057708, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.042065, errors=[]

phase9_baseline_regret_insertion_local_search

- Execution mode: baseline_regret_insertion_local_search.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.451049.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.466391.
- Held-out runtime ms: 1492.19562.
- Robustness dispersion across held-out structure families: 0.086899.
- Worst held-out instances:
- X-n275-k28: feasible=True, penalized gap=0.746105, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.650864, errors=[]
- X-n247-k50: feasible=True, penalized gap=0.395235, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.390341, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.149411, errors=[]

phase9_solver_evolution

- Execution mode: solver_evolution.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 137.1848.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]

- X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

Judge Note

Conservative Summary of Phase-9 CVRP Solver-Evolution Suite

Aspect	Evolved Solver	Baselines (Selected)
Feasibility	100% on held-out conditions	100% for all baselines
Held-out Objective Gap	0.273435 (same as Nearest Neighbor)	Clarke-Wright Savings: best gap 0.073766; Regret Insertion: worse (0.466391)
Runtime (ms)	137.18	Nearest Neighbor: 43.28; Clarke-Wright: 78.47; Regret Insertion: 1492.2
Robustness (dispersion)	0.002777 (low, same as Nearest Neighbor)	Clarke-Wright: 0.007791; Regret Insertion: 0.086899 (higher variability)
Novelty/Complexity	Zero novelty and complexity	Zero for all baselines
Training Feasibility & Gap	100%, gap 0.337704	Matched across baselines

Interpretation

- The evolved solver matches the feasibility and objective gap performance of the nearest neighbor constructive baseline but ****does not improve upon the best baseline (Clarke-Wright Savings)**** in terms of held-out objective gap.
- Runtime for the evolved solver is higher than nearest neighbor and Clarke-Wright but much less than regret insertion.
- Robustness is comparable and low dispersion similar to nearest neighbor.
- No significant novel solver behavior or complexity was introduced.
- Overall, the evolved solver ****does not clearly beat fixed baselines****, especially the Clarke-Wright Savings baseline, which shows the lowest objective gap and competitive runtime.

****Conclusion:**** The phase-9 evolved CVRP solver is feasible and robust but does not outperform the best fixed baseline (Clarke-Wright Savings) in objective gap or runtime on held-out structural families.